🏆 Infosys SE Elite Crack Vault — 02 Percentages Profit Loss Discount and Marked Price Elite Comprehensive Dual-Method Solutions • Mathematical Derivation & 10-Second Exam Shortcuts

Problem 01 • Comparative Percentage in Pie Chart

MUST-SOLVE

Paper cost is approximately what percent of the Printing cost?

A. 125%	B. 120%
C. 80%	D. 115%

✓ **Verified Answer:** A. 125%

Problem 02 • Profit, Loss and Discount

MUST-SOLVE

A man spends 1/5 of his salary on food, 1/10 of his salary on house rent and 3/5 of his salary on clothes. He still has $18000 left with him. Find his salary?

a. 180000	b. 170000
c. 160000	d. 150000

✓ **Verified Answer:** A

📐 DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT

🔷 Method 1: Detailed Pedagogical Derivation (Concept Clarity)

1. Let the unknown total salary be denoted as S.
2. Fractional expenditures given in the problem:
3. Food expenditure: $\frac{1}{5} S$
4. House rent expenditure: $\frac{1}{10} S$
5. Clothes expenditure: $\frac{3}{5} S$
6. Compute the sum of all fractions spent:
   $$\text{Total Fraction Spent} = \frac{1}{5} + \frac{1}{10} + \frac{3}{5} = \frac{2 + 1 + 6}{10} = \frac{9}{10} S$$
7. Compute remaining fraction of salary:
   $$\text{Remaining Balance} = S - \frac{9}{10} S = \frac{1}{10} S$$
8. Equating remaining balance to the given amount of Rs. 18,000:
   $$\frac{1}{10} S = 18,000 \implies S = 18,000 \times 10 = \mathbf{1,80,000}$$

⚡ Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code

- **Percentage Conversion Linking:** $\frac{1}{5} = 20\%$, $\frac{1}{10} = 10\%$, $\frac{3}{5} = 60\%$.
- **Total Spent:** $20\% + 10\% + 60\% = 90\%$.
- **Remaining Percentage:** $100\% - 90\% = 10\% = \frac{1}{10}$.
- **Fast Multiplier:** $10\% = 18,000 \implies 100\% = 18,000 \times 10 = \mathbf{1,80,000}$ (Option A).

Problem 03 • Profit, Loss and Discount MUST-SOLVE

A man spends 20% of his salary on food items, 25% of the remaining on house rent.33.33% of the remaining on entertainment and 10% of the remaining on children's education. After this he pays Rs.900 as income tax and still have Rs.2340. How many Rs. he spends on entertainment?

a. Rs.9000

b. Rs.2800

c. Rs.1800

d. Data inadequate e. None of these

✓ **Verified Answer:** A

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Let the total initial salary be S.
2. After spending 20% on food:
   $$\text{Remaining } R_1 = S \times (1 - 0.20) = 0.80 S = \frac{4}{5} S$$
3. After spending 25% of remainder on house rent:
   $$\text{Remaining } R_2 = R_1 \times (1 - 0.25) = 0.80 S \times 0.75 = 0.60 S = \frac{3}{5} S$$
4. Spends 33.33% ($\frac{1}{3}$) of remainder on entertainment:
   $$\text{Entertainment Expense} = \frac{1}{3} \times R_2 = \frac{1}{3} \times (0.60 S) = 0.20 S = \frac{1}{5} S$$
   $$\text{Remaining } R_3 = R_2 - 0.20 S = 0.40 S = \frac{2}{5} S$$
5. Spends 10% of remainder on education:
   $$\text{Remaining } R_4 = R_3 \times (1 - 0.10) = 0.40 S \times 0.90 = 0.36 S = \frac{9}{25} S$$
6. He pays Rs. 900 income tax and has Rs. 2340 left:
   $$R_4 - 900 = 2340 \implies 0.36 S = 3240 \implies S = \frac{3240}{0.36} = \mathbf{9000} \text{ (Total Salary)}$$
7. Compute Entertainment expenditure:
   $$\text{Entertainment} = 0.20 \times 9000 = \mathbf{Rs. 1800}$$
   (Note: Option A gives Total Salary Rs. 9000; Option C gives Entertainment Expense Rs. 1800).

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- **Telescopic Multiplier Chain:**
  $$S \times \frac{4}{5} \times \frac{3}{4} \times \frac{2}{3} \times \frac{9}{10} = S \times \frac{9}{25} = 0.36 S$$
- **Solve for S in 5 Seconds:**
  $$0.36 S = 2340 + 900 = 3240 \implies S = \frac{3240}{0.36} = \mathbf{9000}$$
- **Entertainment Part:**
  $$9000 \times \left(\frac{4}{5} \times \frac{3}{4} \times \frac{1}{3}\right) = 9000 \times \frac{1}{5} = \mathbf{1800}$$

Problem 04 • Profit, Loss and Discount MUST-SOLVE

A man spends 50% of his income. His income increases by 30% and his expenditure also increase by 20%. Find the Percentage increase in his saving.

a. 20%

b. 30%

c. 40%

d. 50%

✓ **Verified Answer:** C

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Assume the baseline income is 100 units.
2. Given he spends 50% of income:
3. Initial Income = 100
4. Initial Expenditure = 50
5. Initial Savings = Income − Expenditure = $100 - 50 = 50$
6. Apply percentage increments:
7. Income increases by 30%: $\text{New Income} = 100 \times 1.30 = 130$
8. Expenditure increases by 20%: $\text{New Expenditure} = 50 \times 1.20 = 60$
9. Compute new savings:
   $$\text{New Savings} = 130 - 60 = 70$$
10. Compute increase in savings and percentage increase:
    $$\text{Increase} = 70 - 50 = 20$$
    $$\text{Percentage Increase in Savings} = \frac{20}{50} \times 100\% = \mathbf{40\%} \text{ (Option C)}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- **Income-Expenditure Delta Rule:** $\Delta I = \Delta E + \Delta S$.
- $\Delta I = 100 \times 30\% = +30$.
- $\Delta E = 50 \times 20\% = +10$.
- $\implies \Delta S = 30 - 10 = +20$.
- Percentage increase on initial savings of $50 = \frac{20}{50} = \frac{2}{5} = \mathbf{40\%}$ (Instant 3-second mental math).

Problem 05 • Profit, Loss and Discount MUST-SOLVE

A man spends 80% of his income. His income increases by 40% and his expenditure also increase by 25%. Find the Percentage increase in his saving.

a. 100%

b. 150%

c. 200%

d. 250%

✓ **Verified Answer:** A

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Assume baseline income is 100 units.
2. Initial allocation:
3. Initial Income = 100
4. Initial Expenditure = 80 (since he spends 80%)
5. Initial Savings = $100 - 80 = 20$
6. Apply percentage increments:
7. Income increases by 40%: $\text{New Income} = 100 \times 1.40 = 140$
8. Expenditure increases by 25%: $\text{New Expenditure} = 80 \times 1.25 = 100$
9. Compute new savings:
   $$\text{New Savings} = 140 - 100 = 40$$
10. Compute percentage increase in savings:
    $$\text{Increase} = 40 - 20 = 20$$
    $$\text{Percentage Increase} = \frac{20}{20} \times 100\% = \mathbf{100\%} \text{ (Option A)}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- $\Delta I = +40$.
- $\Delta E = 80 \times 25\% = +20$.
- $\Delta S = 40 - 20 = +20$.
- Since base savings is 20, an increase of $+20$ means savings have exactly doubled $\implies \mathbf{100\%}$ increase!

Problem 06 • Percentages

MUST-SOLVE

The price of an article rose by 25% every odd year and fell by 20% every even year, what would be the percentage change after 180 years?

a. 54	b. 69
c. 87	d. 100

✓ **Verified Answer:** D

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Let initial price be P.
2. Odd Year: Rose by 25% $\implies$ Multiplier $= 1 + 0.25 = \frac{5}{4}$.
3. Even Year: Fell by 20% $\implies$ Multiplier $= 1 - 0.20 = \frac{4}{5}$.
4. Combined Multiplier for any 2-year cycle (1 odd + 1 even):
   $$\text{Cycle Multiplier} = \frac{5}{4} \times \frac{4}{5} = 1.00$$
5. In 180 years, there are exactly $180 / 2 = 90$ complete two-year cycles.
6. Price after 180 years $= P \times (1.00)^{90} = P$ (Exact 100% of original value, Option D).

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- $+25\% (\times 1.25) \times -20\% (\times 0.8) = 1.00$.
- Net change per 2-year block is exactly 0%.
- For any even number of years (180 years), the price remains unchanged at $\mathbf{100\%}$ of initial price!

Problem 07 • Percentages

MUST-SOLVE

The population of a town is 18000. It increases by 10% during first year and by 20% during the second year. The population after 2 years will be.....

a. 19800	b. 21600
c. 23760	d. None

✓ **Verified Answer:** C

Problem 08 • Percentages MUST-SOLVE

The length of a rectangle increased by 40% and its breadth increased by 20%. What will be the percentage change in the area of the rectangle?

a. 50% inc

b. 10% inc

c. 44% inc

d. 68% inc e. None of these

✓ **Verified Answer:** D

Problem 09 • Percentages MUST-SOLVE

A student has to secure 40% marks to pass an examination. He secured 75 marks and was declared FAIL 45 marks. Find the maximum marks set for the examination.

a. 600

b. 200

c. 100

d. 300 e. None of these

✓ **Verified Answer:** D

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Let the maximum marks be M.
2. Passing marks required $= 40\%$ of $M = 0.40 M$.
3. The candidate secured 75 marks and failed by 45 marks:
   $$\text{Passing Marks} = 75 + 45 = 120$$
4. Set up equation:
   $$0.40 M = 120 \implies M = \frac{120}{0.40} = \frac{1200}{4} = \mathbf{300}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Required Pass Marks $= 75 + 45 = 120$.
- $40\% = 120 \implies 1\% = \frac{120}{40} = 3$.
- Total Marks (100%) $= 3 \times 100 = \mathbf{300}$ (Option D).

Problem 10 • Percentages MUST-SOLVE

In an examination, the cut off mark is 40% and a student is declared to be failed by 4 marks when he scored 38%. What is the maximum score a student can get in this exam?

a. 100 | b. 300

c. 400 | d. 200 e. None of these

✓ **Verified Answer:** D

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Cut-off passing mark is 40%.
2. The student scored 38% and failed by 4 marks.
3. Difference between pass mark and scored mark:
   $$\Delta \% = 40\% - 38\% = 2\%$$
4. Since 2% corresponds to 4 marks:
   $$2\% \text{ of } M = 4 \implies M = \frac{4}{0.02} = \mathbf{200}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Gap $= 40\% - 38\% = 2\% = 4 \text{ marks}$.
- $1\% = 2 \text{ marks} \implies 100\% = 2 \times 100 = \mathbf{200}$ (Option D).

Problem 11 • Percentages MUST-SOLVE

A student has to secure 30% marks to get through. If he gets 40 marks and fails by 20 marks, find the maximum marks set for the examination.

a. 600	b. 200
c. 100	d. 300 e. None of these

✓ **Verified Answer:** B

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Required passing percentage = 30%.
2. Student scored 40 marks and failed by 20 marks:
   $$\text{Passing Marks} = 40 + 20 = 60$$
3. Equating passing marks to 30%:
   $$30\% \text{ of } M = 60 \implies 0.30 M = 60 \implies M = \frac{60}{0.30} = \mathbf{200}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- $30\% = 60 \implies 10\% = 20 \implies 100\% = \mathbf{200}$ (Option B).

Problem 12 • Percentages MUST-SOLVE

In an examination, P scored 30 % marks and failed by 15 marks. Q scored 40 % marks and obtained 35 marks more than those required to pass. Find the pass percentage.

a. 30%

b. 33%

c. 35%

d. 40%

✓ **Verified Answer:** B

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Let total maximum marks be M.
2. Candidate P: scored 30% and failed by 15 marks:
   $$\text{Passing Marks} = 0.30 M + 15$$
3. Candidate Q: scored 40% and got 35 marks above passing:
   $$\text{Passing Marks} = 0.40 M - 35$$
4. Equating both expressions:
   $$0.40 M - 0.30 M = 35 + 15 \implies 0.10 M = 50 \implies M = 500$$
5. Calculate Passing Marks and Passing Percentage:
   $$\text{Pass Marks} = 0.30(500) + 15 = 150 + 15 = 165$$
   $$\text{Pass Percentage} = \frac{165}{500} \times 100\% = \mathbf{33\%}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Percentage Gap $= 40\% - 30\% = 10\%$.
- Marks Gap $= 35 - (-15) = 50 \text{ marks}$.
- $10\% = 50 \implies 1\% = 5 \text{ marks}$.
- P failed by 15 marks $\implies 15 / 5 = 3\%$.
- Pass Percentage $= 30\% + 3\% = \mathbf{33\%}$ (Option B).

Problem 13 • Percentages MUST-SOLVE

A candidate scoring 25% in an examination fails by 30 marks, while another candidate scores 50 % mark, gets 20 marks more than the minimum pass marks. Find the minimum pass marks. Find the minimum pass percentage.

a. 42%

b. 55%

c. 33%

d. 40%

✓ **Verified Answer:** D

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Let maximum marks be M.
2. From first candidate: $\text{Pass Marks} = 0.25 M + 30$.
3. From second candidate: $\text{Pass Marks} = 0.50 M - 20$.
4. Equating both:
   $$0.50 M - 0.25 M = 30 + 20 \implies 0.25 M = 50 \implies M = 200$$
5. Compute passing marks and percentage:
   $$\text{Pass Marks} = 0.25(200) + 30 = 50 + 30 = 80$$
   $$\text{Pass Percentage} = \frac{80}{200} \times 100\% = \mathbf{40\%}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Gap: $50\% - 25\% = 25\% = 20 + 30 = 50 \text{ marks}$.
- $25\% = 50 \implies 1\% = 2 \text{ marks}$.
- Failure of 30 marks $= 30 / 2 = 15\%$.
- Pass Percentage $= 25\% + 15\% = \mathbf{40\%}$ (Option D).

Problem 14 • Percentages MUST-SOLVE

In a test consisting of 300 questions, Deepika answered 40% of the first 100 questions correctly. What per cent of the remaining 200 questions does she need to answer correctly for her grade on the entire exam to be 50%?

a. 75%	b. 55%
c. 60%	d. Cannot be determined e. None of these

✓ **Verified Answer:** B

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Total questions in exam = 300.
2. Target grade = 50% of 300:
   $$\text{Target Correct Questions} = 0.50 \times 300 = 150$$
3. First 100 questions: answered 40% correctly:
   $$\text{Correct so far} = 0.40 \times 100 = 40$$
4. Questions still needed from remaining 200 questions:
   $$\text{Needed} = 150 - 40 = 110$$
5. Required percentage on remaining 200 questions:
   $$\text{Percentage} = \frac{110}{200} \times 100\% = \mathbf{55\%}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- **Weighted Average / Alligation:**
- Part 1 (100 Q, Weight 1) has 40%.
- Part 2 (200 Q, Weight 2) has $x\%$.
- Combined Target has 50%.
- $1(40) + 2(x) = 3(50) \implies 40 + 2x = 150 \implies 2x = 110 \implies x = \mathbf{55\%}$ (Option B).

Problem 15 • Percentages MUST-SOLVE

There were only two candidates in an election. A candidate got 62% votes and he won by 144 votes. How many votes polled?

a. 600

b. 6000

c. 500

d. None

✓ **Verified Answer:** A

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Let total votes polled be V.
2. Winner obtained 62% of votes:
   $$\text{Winner} = 62\%$$
   $$\text{Loser} = 100\% - 62\% = 38\%$$
3. Margin of victory:
   $$\text{Margin} = 62\% - 38\% = 24\%$$
4. Since winner won by 144 votes:
   $$0.24 V = 144 \implies V = \frac{144}{0.24} = \mathbf{600}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Victory Gap $= 62\% - 38\% = 24\%$.
- $24\% = 144 \implies 1\% = \frac{144}{24} = 6$.
- Total Votes (100%) $= 6 \times 100 = \mathbf{600}$ (Option A).

Problem 16 • Percentages MUST-SOLVE

In an election between two candidates, the winner secured 58% of the total votes cast and wins by a majority of 2400 votes. How many votes polled?

a. 15000

b. 16978

c. 17000

d. 18960

✓ **Verified Answer:** A

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Let total votes cast be V.
2. Winner obtained 58% of votes:
   $$\text{Winner} = 58\% \implies \text{Loser} = 100\% - 58\% = 42\%$$
3. Margin of majority:
   $$\Delta \% = 58\% - 42\% = 16\%$$
4. Equating to given majority of 2400 votes:
   $$0.16 V = 2400 \implies V = \frac{2400}{0.16} = \mathbf{15,000}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Majority Gap $= 58\% - 42\% = 16\%$.
- $16\% = 2400 \implies 1\% = \frac{2400}{16} = 150$.
- Total Votes (100%) $= 150 \times 100 = \mathbf{15,000}$ (Option A).

Problem 17 • Percentages MUST-SOLVE

In an election between two candidates, one gets 72% of the total votes. If the total votes are 8200, by how many votes did the winner win the election?

a. 1835

b. 1722

c. 3608

d. 4428 e. None of these

✓ **Verified Answer:** C

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Total votes = 8200.
2. Winner's share = 72% $\implies$ Loser's share = $100\% - 72\% = 28\%$.
3. Difference between winner and loser:
   $$\Delta \% = 72\% - 28\% = 44\%$$
4. Winning margin in votes:
   $$\text{Margin} = 0.44 \times 8200 = 44 \times 82 = \mathbf{3608}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Fast multiplication: $44 \times 82 = 44 \times (80 + 2) = 3520 + 88 = \mathbf{3608}$ (Option C).

Problem 18 • Percentages MUST-SOLVE

In a college election between two students, 10% of the votes cast are invalid. The winner gets 70% of the valid votes and defeats the loser by 1800 votes. How many votes were totally cast?

a. 1800

b. 7200

c. 5000

d. 3600 e. None of these

✓ **Verified Answer:** C

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Let total votes cast be V.
2. 10% votes are invalid $\implies$ Valid votes $= 90\%$ of $V = 0.90 V$.
3. Winner gets 70% of valid votes $\implies$ Loser gets 30% of valid votes.
4. Margin of victory $= 70\% - 30\% = 40\%$ of valid votes.
5. Equating margin to 1800 votes:
   $$0.40 \times (0.90 V) = 1800 \implies 0.36 V = 1800 \implies V = \frac{1800}{0.36} = \mathbf{5000}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- $V \times \frac{9}{10} \times \frac{4}{10} = 0.36 V$.
- $0.36 V = 1800 \implies V = \frac{1800}{0.36} = \mathbf{5000}$ (Option C).

Problem 19 • Profit, Loss and Discount MUST-SOLVE

A speaks truth in 60% of the cases, while B in 90% of the cases. In what percent of cases are they likely to contradict each other in stating the same fact?

a. 80%

b. 78%

c. 55%

d. 42%

✓ **Verified Answer:** D

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Probability of truth and lie for each person:
2. $P(A) = 0.60, P(A') = 1 - 0.60 = 0.40$
3. $P(B) = 0.90, P(B') = 1 - 0.90 = 0.10$
4. Contradiction occurs when one speaks truth and the other lies:
   $$\text{Contradiction} = (A \text{ speaks truth } \cap B \text{ lies}) \cup (A \text{ lies } \cap B \text{ speaks truth})$$
5. Compute probabilities:
   $$P = (0.60 \times 0.10) + (0.40 \times 0.90) = 0.06 + 0.36 = 0.42 = \mathbf{42\%}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Cross-product sum: $(60\% \times 10\%) + (40\% \times 90\%) = 6\% + 36\% = \mathbf{42\%}$ (Option D).

Problem 20 • Percentages MUST-SOLVE

A Sum was put at SI at a certain rate for 2 years. Had it been put at 3% higher rate, it would have fetched Rs.300 more. Find the sum.

a. 2500 | b. 5000

c. 7769 | d. 9697

✓ **Verified Answer:** B

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Simple Interest formula: $SI = \frac{P \times R \times T}{100}$.
2. Increase in interest rate $= 3\%$. Time $T = 2$ years.
3. Extra interest fetched:
   $$\Delta SI = \frac{P \times 3 \times 2}{100} = \frac{6P}{100}$$
4. Equating extra interest to Rs. 300:
   $$\frac{6P}{100} = 300 \implies P = \frac{300 \times 100}{6} = 50 \times 100 = \mathbf{5000}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Effective extra interest rate over 2 years $= 3\% \times 2 = 6\%$.
- $6\% = 300 \implies 1\% = 50 \implies 100\% = \mathbf{5000}$ (Option B).

Problem 21 • Percentages MUST-SOLVE

10% of the voters did not cast their votes in an election between two candidates 10% of the votes polled were found invalid. The successful candidate got 54% of the valid votes and won by a margin of 1620 votes. The number of voters enrolled in the voter's list was

a. 25000 | b. 27000

c. 30000 | d. 34000

✓ **Verified Answer:** A

Problem 22 • Percentages MUST-SOLVE

In a college election between two students, 10% of the votes cast are invalid. The winner gets 70% of the valid votes and defeats the loser by 1800 votes. How many votes were totally cast?

a. 1800
b. 7200
c. 5000
d. 3600 e. None of these

✓ **Verified Answer:** C

Problem 23 • Percentages MUST-SOLVE

In a election between two candidate 20% did not vote & 12(1/2)% i.e 12.5 %of votes were invalid winner got 60% of valid votes and was elected by a majority of 5600 votes. Find the total number of votes?

a. 40000
b. 45000
c. 30000
d. NOT

✓ **Verified Answer:** A

Problem 24 • Profit, Loss and Discount MUST-SOLVE

On selling a pen at 5% loss and a book at 15% gain, Karim gains Rs.7. If he sells the pen at 5% gain and the book at 10% gain, he gains Rs.13. What is the cost price of the book in Rupees?

a. 80

b. 85

c. 100

d. 95

✓ **Verified Answer:** A

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Let cost price of pen $= P$, cost price of book $= B$.
2. Condition 1: 5% loss on pen and 15% gain on book gives Rs. 7 gain:
   $$-0.05 P + 0.15 B = 7 \quad \text{--- (Eq 1)}$$
3. Condition 2: 5% gain on pen and 10% gain on book gives Rs. 13 gain:
   $$+0.05 P + 0.10 B = 13 \quad \text{--- (Eq 2)}$$
4. Adding Eq 1 and Eq 2 eliminates P completely:
   $$(-0.05 P + 0.05 P) + (0.15 B + 0.10 B) = 7 + 13$$
   $$0.25 B = 20 \implies B = \frac{20}{0.25} = 20 \times 4 = \mathbf{80}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Directly add both transactions: Pen terms cancel out instantly ($-5\% + 5\% = 0$).
- Total book gain $= 15\% + 10\% = 25\% = \frac{1}{4}$.
- $\frac{1}{4} B = 7 + 13 = 20 \implies B = 20 \times 4 = \mathbf{80}$ (Option A).

Problem 25 • Percentages MUST-SOLVE

If A's salary is 25% less than B's salary, then B's salary is what % more than A's salary?

a. Cannot be determined	b. 21
c. 33.33	d. 25 e. 20

✓ **Verified Answer:** C

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Let B's salary be 100.
2. A's salary is 25% less: $A = 100 - 25 = 75$.
3. Difference between B and A: $100 - 75 = 25$.
4. Percentage by which B is more than A:
   $$\frac{B - A}{A} \times 100\% = \frac{25}{75} \times 100\% = \frac{1}{3} \times 100\% = \mathbf{33.33\%}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- **Fraction Linking Invariance:** A decrease of $\frac{1}{n}$ requires an increase of $\frac{1}{n-1}$.
- Decrease of $25\% = \frac{1}{4} \implies$ Increase of $\frac{1}{4-1} = \frac{1}{3} = \mathbf{33.33\%}$ (Option C).

Problem 26 • Percentages MUST-SOLVE

Study the following table and answer the
Revenue (in INR lakhs) of a company ABC from
the sale of its 4 product P, Q, R and S 6 years
Prod
uct /
Year
20
13
20
14
20
15
20
16
20
17
20
18
P
16
4
16
8
17
4
17
5
19
2
18
0
Q
15
7
16
4
17
6
16
7
17
9
18
4
R
18
3
17
7
18

6
17
3
18
6
18
8
S
17
6
17
1
19
2
18
0
19
0
20
6

The total revenue of the company from the sale of product Q in 2015 and 218 and from the sale of product S in 2015 is approximately what percentage less than the total revenue from the sale of product P in 2014 and 2017 and product R in 2013 and 2014?

a. 23.33
b. 19.33
c. 15.33
d. 11.33

✓ **Verified Answer:** A

Problem 27 • Percentages MUST-SOLVE

In a company 58% of the employees are males and rest are females 48% of the employees are graduates of which 33
1
3% are males, among
females the percentage of those who are not graduates is _____ (Correct to one decimal place)

a. 25.2
b. 23.8
c. 30.6
d. 33.3

✓ **Verified Answer:** B

Problem 28 • Percentages MUST-SOLVE

The income of A is 70% of the income B and the expenditure of B is 20% less than the expenditure of A. if the expenditure of A is equal to 60% of B's income, what is the ratio of saving of A and B?

a. 5:26

b. 7:13

c. 7:26

d. 5:13

✓ **Verified Answer:** A

Problem 29 • Profit, Loss and Discount MUST-SOLVE

₹6200 divided into three parts proportional to
1
2 :
1
3 :
1
5 and respectively

a. 1000, 2000, 3001

b. 2000, 3000, 4045

c. 3000, 2000, 1200

d. 4000, 5000, 6978

✓ **Verified Answer:** C

Problem 30 • Profit, Loss and Discount

MUST-SOLVE

A bag contains an equal number of one rupee 50 paise and 25 paise coins respectively. If the total value is Rs.35, how many coins of each type are there?

a. 10

b. 20

c. 30

d. 40

✓ **Verified Answer:** B

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Let the number of coins of each denomination be x.
2. Denominations in Rupees:
3. 1 Rupee = 1.00
4. 50 Paise = 0.50
5. 25 Paise = 0.25
6. Total value of 1 set of coins $= 1.00 + 0.50 + 0.25 = 1.75$ Rupees.
7. Total value equation:
   $$1.75 x = 35 \implies x = \frac{35}{1.75} = \frac{3500}{175} = \mathbf{20}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- 1 Rupee + 50p + 25p $= 1.75$ Rs.
- Number of coins of each type $= \frac{35}{1.75} = \frac{35}{7/4} = 35 \times \frac{4}{7} = 5 \times 4 = \mathbf{20}$ (Option B).

Problem 31 • Percentages

MUST-SOLVE

A sum of x is divided between A,B,C and D such that the ratio of share of A and B is 9:5, C's share is 70% of B's share and the ratio of the share of D to the combined share of B and C is 1:3. If the share of A is INR 999, the value of x is

a. 4478

b. 2257

c. 2239

d. 4514

✓ **Verified Answer:** B

Problem 32 • Profit, Loss and Discount

MUST-SOLVE

The profit that a person makes on selling 9 apples is equal to the cost of 2 apples. What is the percent of profit?

a. 11.11%	b. 22.22%
c. 27.27%	d. NOT

✓ **Verified Answer:** B

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Let the Cost Price of 1 apple be CP.
2. Cost price of 9 apples $= 9 CP$.
3. Profit made on selling 9 apples $= 2 CP$.
4. Profit percentage formula:
   $$\text{Profit } \% = \frac{\text{Profit}}{\text{Cost Price}} \times 100\% = \frac{2 CP}{9 CP} \times 100\% = \frac{2}{9} \times 100\% = \mathbf{22.22\%}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Fraction Profit $= \frac{2}{9}$.
- From Fraction Cheat Table: $\frac{1}{9} = 11.11\% \implies \frac{2}{9} = 2 \times 11.11\% = \mathbf{22.22\%}$ (Option B).

Problem 33 • Profit, Loss and Discount

MUST-SOLVE

Renu bought 28 articles for INR 1540. She sold them at a profit equal to the selling price of 6 articles. How many articles did she sell for INR 840?

a. 15	b. 12
c. 10	d. 14

✓ **Verified Answer:** B

Problem 34 • Percentages MUST-SOLVE

The ratio in which two sugar solutions of the concentration 15% and 40% are to be mixed to get a solution of concentration 30% is:

a. 2:3

b. 3:2

c. 8:9

d. 9:8

✓ **Verified Answer:** A

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Let quantity of 15% solution be x and 40% solution be y.
2. Resulting mixture concentration is 30%:
   $$0.15 x + 0.40 y = 0.30 (x + y)$$
3. Rearranging terms:
   $$0.40 y - 0.30 y = 0.30 x - 0.15 x \implies 0.10 y = 0.15 x$$
4. Ratio of $x : y$:
   $$\frac{x}{y} = \frac{0.10}{0.15} = \frac{2}{3} = \mathbf{2 : 3}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- **Alligation Cross:**
- Cheaper (15%) vs Dearer (40%) around Mean (30%).
- Ratio $= (40 - 30) : (30 - 15) = 10 : 15 = \mathbf{2 : 3}$ (Option A).

Problem 35 • Profit, Loss and Discount MUST-SOLVE

Vessels A and B contain mixtures of milk and water in the ratios 4:5 and 5:1 respectively. In what ratio should quantities of mixtures be taken from A and B to form a mixture in which milk to water is in the ratio 5:4?

a. 5:5

b. 5:4

c. 5:3

d. 5:2

✓ **Verified Answer:** D

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Fraction of milk in Vessel A (ratio 4:5): $\text{Milk}_A = \frac{4}{4+5} = \frac{4}{9}$.
2. Fraction of milk in Vessel B (ratio 5:1): $\text{Milk}_B = \frac{5}{5+1} = \frac{5}{6}$.
3. Desired fraction of milk in final mixture (ratio 5:4): $\text{Milk}_M = \frac{5}{5+4} = \frac{5}{9}$.
4. Let ratio of mixtures taken be $x : y$:
   $$\frac{4}{9} x + \frac{5}{6} y = \frac{5}{9} (x + y)$$
   $$\left(\frac{5}{6} - \frac{5}{9}\right) y = \left(\frac{5}{9} - \frac{4}{9}\right) x \implies \frac{5}{18} y = \frac{1}{9} x = \frac{2}{18} x \implies \frac{x}{y} = \frac{5}{2}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Express all fractions with common denominator 18:
- $A = \frac{8}{18}$, $B = \frac{15}{18}$, Mean $= \frac{10}{18}$.
- Cross difference: $(15 - 10) : (10 - 8) = \mathbf{5 : 2}$ (Option D).

Problem 36 • Profit, Loss and Discount

MUST-SOLVE

A person bought 2 varieties of tea, 1st at Rs.40/kg and 2nd at Rs.70/kg. He mixed both varieties and sold the mixture at Rs.55/kg making a profit of 10%. In what ratio did he mixed the two varieties of tea?

a. 1:2

b. 2:3

c. 3:2

d. 2:1

✓ **Verified Answer:** D

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Selling price of mixture = Rs. 55 with 10% profit.
2. Calculate Mean Cost Price (CP_M) of the mixture:
   $$CP_M = \frac{55}{1 + 0.10} = \frac{55}{1.10} = \mathbf{50 \text{ Rs/kg}}$$
3. Cost price of Variety 1 = Rs. 40/kg; Cost price of Variety 2 = Rs. 70/kg.
4. Apply Rule of Alligation on Cost Price:
   $$\text{Ratio} = \frac{70 - 50}{50 - 40} = \frac{20}{10} = \mathbf{2 : 1}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Multiplier: $CP = \frac{55}{1.1} = 50$.
- Cross: $|70 - 50| : |50 - 40| = 20 : 10 = \mathbf{2 : 1}$ (Option D).

Problem 37 • Percentages

MUST-SOLVE

300 grams of sugar solution has 40% of sugar in it. How much sugar should be added to make it 50% in the solution?

a. 10 gms

b. 60 gms

c. 40 gms

d. 80 gms

✓ **Verified Answer:** B

Problem 38 • Percentages MUST-SOLVE

In 50 gram of alloy gold and silver. The gold is 80% by weight. How much gold should be mixed to this alloy so that the weight of gold would become 95%?

a. 100

b. 120

c. 130

d. 150

✓ **Verified Answer:** D

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Total initial alloy weight = 50 grams.
2. Initial composition:
3. Gold = 80% of 50g = 40 grams.
4. Silver = 50g − 40g = 10 grams.
5. Pure gold is added $\implies$ weight of Silver remains invariant at 10 grams.
6. In the new alloy, Gold is 95% $\implies$ Silver must be $100\% - 95\% = 5\%$.
7. Let new total weight be W:
   $$0.05 W = 10 \implies W = \frac{10}{0.05} = 200 \text{ grams}$$
8. Amount of Gold added:
   $$\text{Added Gold} = 200 - 50 = \mathbf{150 \text{ grams}}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- **Invariant Component Rule:**
- Initial: Silver $= 20\% \times 50 = 10\text{g}$.
- Target: Silver $= 5\% = \frac{1}{20}$ of new total.
- New total $= 10 \times 20 = 200\text{g} \implies$ Added $= 200 - 50 = \mathbf{150\text{g}}$ (Option D).

Problem 39 • Percentages MUST-SOLVE

A mixture of 40 litres of milk and water contains 10% of water. How much water is to be added to the mixture so that the water may be 20% in the new mixture.

a. 5 litres

b. 6 litres

c. 7 litres

d. 8 litres

✓ **Verified Answer:** A

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Total volume = 40 litres.
2. Initial composition:
3. Water = 10% of 40 = 4 litres.
4. Milk = 40 − 4 = 36 litres.
5. Water is added $\implies$ Milk remains constant at 36 litres.
6. In new mixture, Water is 20% $\implies$ Milk is 80%.
7. Let new total volume be V:
   $$0.80 V = 36 \implies V = \frac{36}{0.80} = 45 \text{ litres}$$
8. Water added:
   $$\text{Added Water} = 45 - 40 = \mathbf{5 \text{ litres}}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Milk invariant: $90\% \times 40 = 80\% \times V \implies V = \frac{36}{0.8} = 45$.
- Water added $= 45 - 40 = \mathbf{5\text{L}}$ (Option A).

Problem 40 • Profit, Loss and Discount

MUST-SOLVE

A starts business with Rs.3500 and after 5 months, B joins with A as his partner. After a year, the profit is divided in the ratio 2 : 3. What is B's contribution in the capital?

a. Rs.7500

b. Rs.8000

c. Rs.8500

d. Rs.9000

✓ **Verified Answer:** D

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Profit sharing ratio equals the ratio of effective investments:
   $$\text{Profit Ratio} = (C_A \times T_A) : (C_B \times T_B)$$
2. A invests Rs. 3500 for the full year (12 months):
   $$\text{Effective Investment}_A = 3500 \times 12 = 42,000$$
3. B joins after 5 months, remaining for $12 - 5 = 7$ months with capital C_B:
   $$\text{Effective Investment}_B = 7 \times C_B$$
4. Given profit ratio $= 2 : 3$:
   $$\frac{42000}{7 C_B} = \frac{2}{3} \implies \frac{6000}{C_B} = \frac{2}{3} \implies 2 C_B = 18000 \implies C_B = \mathbf{9000}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- $\frac{3500 \times 12}{7 \times C_B} = \frac{2}{3} \implies \frac{500 \times 12}{C_B} = \frac{2}{3} \implies C_B = 3000 \times 3 = \mathbf{9000}$ (Option D).

Problem 41 • Percentages

MUST-SOLVE

5%
b.
18%
c.
14%
d.
10%
e.
8%

b. 18%

c. 14%

d. 10% e. 8%

✓ **Verified Answer:** A

Problem 42 • Profit, Loss and Discount MUST-SOLVE

How much more is spent on Hockey than on Basketball?

a. 15cr | b. 12cr

c. 18cr | d. 20cr e. 10cr

✓ **Verified Answer:** C

Problem 43 • Profit, Loss and Discount MUST-SOLVE

If total spending on Hockey is increased by 20% while that of football is decreased by 30%, what will be different between the spending on hockey and football now?

a. 11.8cr | b. 11.2cr

c. 10.9cr | d. 12.1cr e. 12.8cr

✓ **Verified Answer:** B

Problem 44 • Profit, Loss and Discount MUST-SOLVE

Study the following graph carefully to answer the questions given below.
Number of Students in different grades in three schools in 2015

In year 2016 the number of students of class XII in three schools A, B and C increase by 5%, 10% and 20% respectively with comparison to the last year in same class. Find the ratio of students in class XII of all schools in 2016?

a. 84 : 78 : 77 | b. 84 : 77 : 78

c. 88 : 77 : 78 | d. 8 : 7 : 9 e. 78 : 84 : 77

✓ **Verified Answer:** B

Problem 45 • Profit, Loss and Discount MUST-SOLVE

Directions (42-46): Study the following information to answer the give questions.
In a certain code
For profit order now
is written as 'ho
ja ye ga',
Right now for him is written as 'ga ve ja se',
Place order for profit
is written as 'ga
bi ho ye' and
Only in right order is written as 've du ye zo'

What is the code for 'for'?
What does 'bi' stand for?

a. Profit
b. Order
c. Place
d. For e. Now

✓ **Verified Answer:** D

Problem 46 • Profit, Loss and Discount MUST-SOLVE

The length breadth and height of a cuboid are increased by 10%, 10% and 20% respectively. Find the percentage increase in its volume.

a. 45.2%

b. 46.2%

c. 47.2%

d. 48.2%

✓ **Verified Answer:** A

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Volume of cuboid $= L \times B \times H$.
2. Given percentage increases:
3. Length: $+10\% \implies 1.10 L$
4. Breadth: $+10\% \implies 1.10 B$
5. Height: $+20\% \implies 1.20 H$
6. New Volume:
   $$V' = 1.10 L \times 1.10 B \times 1.20 H = (1.10 \times 1.10 \times 1.20) \times (L B H)$$
7. Calculate composite multiplier:
   $$1.10 \times 1.10 = 1.21$$
   $$1.21 \times 1.20 = 1.452$$
8. Net Percentage Increase:
   $$\text{Percentage Increase} = (1.452 - 1) \times 100\% = \mathbf{45.2\%}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Successive Multiplier: $1.1 \times 1.1 \times 1.2 = 1.21 \times 1.2 = 1.452$.
- Net change $= +45.2\%$ (Option A).

Problem 47 • Profit, Loss and Discount MUST-SOLVE

A ship 55 km from the shore springs a leak which admits 2 tonnes of water in 6 minutes, 80 tonnes would suffice to sink her, but the pumps can throw out 12 tonnes an hour. The average rate of sailing that she may just reach the shore as she begins to sink is

a. 9.17 km/h

b. 0.97 km/h

c. 55 km/h

d. 5.5 km/h e. None of these

✓ **Verified Answer:** D

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Water inflow rate:
   $$\text{Inflow} = \frac{2 \text{ tonnes}}{6 \text{ minutes}} = \frac{2}{1/10 \text{ hr}} = 20 \text{ tonnes/hour}$$
2. Water pumping outflow rate:
   $$\text{Outflow} = 12 \text{ tonnes/hour}$$
3. Net accumulation rate:
   $$\text{Net Rate} = 20 - 12 = 8 \text{ tonnes/hour}$$
4. Capacity to sink is 80 tonnes. Time available before ship sinks:
   $$\text{Time} = \frac{80 \text{ tonnes}}{8 \text{ tonnes/hr}} = 10 \text{ hours}$$
5. Distance to shore is 55 km. Required sailing speed:
   $$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{55 \text{ km}}{10 \text{ hours}} = \mathbf{5.5 \text{ km/h}}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Net Inflow $= 20 - 12 = 8$ tonnes/hr.
- Time to sink $= 80 / 8 = 10$ hrs.
- Sailing speed needed $= 55 / 10 = \mathbf{5.5 \text{ km/h}}$ (Option D).

Problem 48 • Profit, Loss and Discount MUST-SOLVE

Anna sold his car Boney at a profit of 20% and Boney sold it to Chakori at a profit of 10%. Chakori sold it to mechanic at a loss of 9.09%. Mechanic spent 10% of his purchasing price and then sold it at a profit of 8.33% to Anna once again. What is the loss of Anna?

a. 23%

b. 43%

c. 50%

d. 40%

✓ **Verified Answer:** A

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Assume Anna's original car purchase price was 100.
2. Anna sells to Boney at 20% profit:
   $$\text{Selling Price}_1 = 100 \times 1.20 = 120 \quad \text{(Anna receives 120)}$$
3. Boney sells to Chakori at 10% profit:
   $$\text{Price to Chakori} = 120 \times 1.10 = 132$$
4. Chakori sells to mechanic at 9.09% ($\frac{1}{11}$) loss:
   $$\text{Price to Mechanic} = 132 \times \left(1 - \frac{1}{11}\right) = 132 \times \frac{10}{11} = 120$$
5. Mechanic spends 10% on repairs and sells at 8.33% ($\frac{1}{12}$) profit:
   $$\text{Mechanic's Total Cost} = 120 \times 1.10 = 132$$
   $$\text{Price Anna Pays to Buy Back} = 132 \times \left(1 + \frac{1}{12}\right) = 132 \times \frac{13}{12} = 143$$
6. Compute Anna's net balance:
   $$\text{Net Balance} = 120 \text{ (received)} - 143 \text{ (paid)} = -23$$
   $$\text{Anna's Net Loss} = \mathbf{23\%}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Multiplier track: $100 \xrightarrow{+20\%} 120 \xrightarrow{+10\%} 132 \xrightarrow{-1/11} 120 \xrightarrow{+10\%} 132 \xrightarrow{+1/12} 143$.
- Net Loss $= 143 - 120 = \mathbf{23\%}$ (Option A).

Problem 49 • Profit, Loss and Discount MUST-SOLVE

In a company 20% of the employees can speak in Hindi only, 40% of the remaining employees can speak in English only. The remaining 2880 employees can speak in both English and Hindi. Find the total number of employees in the company.

a. 4500

b. 6000

c. 6500

d. 5200

✓ **Verified Answer:** B

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Let total employees be E.
2. 20% speak Hindi only $\implies$ Remaining $= 80\%$ of $E = 0.80 E$.
3. 40% of remaining speak English only:
   $$\text{English only} = 0.40 \times 0.80 E = 0.32 E$$
4. Remaining speak both languages:
   $$\text{Both} = 0.80 E - 0.32 E = 0.48 E$$
5. Given both languages count is 2880:
   $$0.48 E = 2880 \implies E = \frac{2880}{0.48} = \frac{288000}{48} = \mathbf{6000}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Multipliers: $E \times (1 - 0.20) \times (1 - 0.40) = E \times 0.8 \times 0.6 = 0.48 E$.
- $0.48 E = 2880 \implies E = \frac{2880}{0.48} = \mathbf{6000}$ (Option B).

Problem 50 • Profit, Loss and Discount MUST-SOLVE

Amit started a business on 1st January 2018, with an investment of Rs.95000. Raghav joined him after few months by investing Rs.76000. If the total profit is divided in the ratio of 5:3 on 1st January 2019, find the period after which Raghav joined the business

a. 1 Month

b. 3 Month

c. 2 Month

d. 4 Month

✓ **Verified Answer:** B

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Amit invests Rs. 95,000 for the full 12 months.
2. Raghav invests Rs. 76,000 for t months.
3. Profit sharing ratio given is $5 : 3$:
   $$\frac{95000 \times 12}{76000 \times t} = \frac{5}{3}$$
4. Simplify numbers: $\frac{95}{76} = \frac{19 \times 5}{19 \times 4} = \frac{5}{4}$.
   $$\frac{5 \times 12}{4 \times t} = \frac{5}{3} \implies \frac{15}{t} = \frac{5}{3} \implies 5t = 45 \implies t = 9 \text{ months}$$
5. Raghav remained in business for 9 months $\implies$ he joined after:
   $$\text{Period} = 12 - 9 = \mathbf{3 \text{ months}}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- Investment ratio: $95 : 76 = 5 : 4$.
- Profit ratio: $5 : 3$.
- Time ratio: $\frac{5/5}{3/4} = \frac{1}{3/4} = \frac{4}{3} \implies \text{Time}_A : \text{Time}_R = 4 : 3$.
- Since A is 12 months (4 units $\implies 1 \text{ unit} = 3$), Raghav's time $= 3 \times 3 = 9$ months.
- Joined after $12 - 9 = \mathbf{3 \text{ months}}$ (Option B).

Problem 51 • Profit, Loss and Discount MUST-SOLVE

A shopkeeper purchased 60 pencil boxes for Rs.4500 and sold at the rate of Rs.4125 for 50 pencil boxes. How much profit present does he make on selling all the pencil boxes?

a. 5%

b. 12%

c. 7%

d. 10%

✓ **Verified Answer:** D

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Total purchase: 60 boxes for Rs. 4500:
   $$\text{Cost Price of 1 box} = \frac{4500}{60} = \text{Rs. } 75$$
2. Sale: 50 boxes for Rs. 4125:
   $$\text{Selling Price of 1 box} = \frac{4125}{50} = \text{Rs. } 82.50$$
3. Profit per box:
   $$\text{Profit} = 82.50 - 75 = \text{Rs. } 7.50$$
4. Profit percentage:
   $$\text{Profit } \% = \frac{7.50}{75} \times 100\% = \frac{1}{10} \times 100\% = \mathbf{10\%}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- $CP = 75$, $SP = 82.50$.
- Multiplier $= \frac{82.5}{75} = 1.10 \implies \mathbf{10\%}$ Profit (Option D).

Problem 52 • Profit, Loss and Discount MUST-SOLVE

Sneha self an article for Rs.864 after pricing 28% decreased on the labelled price and make 44% profit on the cost price. What would have been the percentage profit on the cost price, if she had sold the article on 20% discount instead of 28%?

a. 59%
b. 60%
c. 45%
d. 50%

✓ **Verified Answer:** B

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Selling price $SP_1 = 864$ with 28% discount on Marked Price (MP):
   $$MP \times (1 - 0.28) = 864 \implies 0.72\ MP = 864 \implies MP = \frac{864}{0.72} = \text{Rs. } 1200$$
2. Also, $SP_1 = 864$ yields 44% profit on Cost Price (CP):
   $$CP \times (1 + 0.44) = 864 \implies 1.44\ CP = 864 \implies CP = \frac{864}{1.44} = \text{Rs. } 600$$
3. If sold at 20% discount on Marked Price:
   $$\text{New } SP_2 = 1200 \times (1 - 0.20) = 1200 \times 0.80 = \text{Rs. } 960$$
4. Compute profit on Cost Price:
   $$\text{Profit} = 960 - 600 = \text{Rs. } 360$$
   $$\text{Percentage Profit} = \frac{360}{600} \times 100\% = \mathbf{60\%}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- **Relative Ratio Shortcut:**
  $$\frac{SP_2}{SP_1} = \frac{100 - 20}{100 - 28} = \frac{80}{72} = \frac{10}{9}$$
- Since SP_1 represents 144% of CP:
  $$\text{New } SP_2\ \% = 144\% \times \frac{10}{9} = 16\% \times 10 = 160\%$$
- Profit $= 160\% - 100\% = \mathbf{60\%}$ (Option B, zero calculation of actual Rupees!).

Problem 53 • Percentages MUST-SOLVE

What will be the central angle for passenger travelling from Patna to Buxar if it is put in the form of a pie-chart?
Total number of passenger travelling from Patna Junction to different districts = 12000

a. 57.6°
b. 57.7°
c. 57.8°
d. 57.9°

✓ **Verified Answer:** A

Problem 54 • Percentages MUST-SOLVE

The number of passengers travelling from Patna to Buxar is what cent more than the number of passengers travelling to Jehanabad?
Total number of passengers travelling from Patna Junction to different districts = 12000

a. 30.33%
b. 31.33%
c. 32.33%
d. 33.33%

✓ **Verified Answer:** D

Problem 55 • Percentages MUST-SOLVE

The number of passengers travelling from Patna to Gaya is what per cent of the total number of passengers travelling from Patna to Darbhanga and Ara together?
Total number of passengers travelling from Patna Junction to different districts = 12000

a. 65%
b. 70%
c. 75%
d. 80%

✓ **Verified Answer:** C

Problem 56 • Percentages MUST-SOLVE

The number of girls appearing for an admission test is twice the number of boys. If 30% of the girls and 45% of the boys get admission, the percentage of candidates who do not get admission is

a. 35 | b. 50

c. 60 | d. 65

✓ **Verified Answer:** D

📐 **DUAL-METHOD SOLUTION: MATHEMATICAL DERIVATION & 10-SECOND EXAM SHORTCUT**

🔷 **Method 1: Detailed Pedagogical Derivation (Concept Clarity)**

1. Assume the number of boys is 100.
2. Given number of girls is twice the number of boys:
   $$\text{Boys} = 100, \quad \text{Girls} = 200, \quad \text{Total Candidates} = 300$$
3. Compute admissions granted:
4. Girls admitted = 30% of 200 = 60
5. Boys admitted = 45% of 100 = 45
6. Total candidates admitted = $60 + 45 = 105$
7. Candidates not admitted:
   $$\text{Not Admitted} = 300 - 105 = 195$$
8. Percentage of candidates who do not get admission:
   $$\text{Percentage} = \frac{195}{300} \times 100\% = \frac{195}{3} = \mathbf{65\%}$$

⚡ **Method 2: 10-Second Exam Shortcut / Multiplier Cheat Code**

- **Weighted Average of Rejections:**
- Boys rejection rate $= 100\% - 45\% = 55\%$ (Weight = 1).
- Girls rejection rate $= 100\% - 30\% = 70\%$ (Weight = 2).
- Overall rejection rate $= \frac{1(55) + 2(70)}{1 + 2} = \frac{55 + 140}{3} = \frac{195}{3} = \mathbf{65\%}$ (Option D).